



Temporal changes of Solid Waste at Limestone Quarries in and Around Yerraguntla, Y.S.R District, A.P using Google Earth Images

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ABSTRACT In lime stone mining and cement industry, wastes are generated in every stage of the operations and are required to handle properly. The types of waste generated from both the industries are solid, liquid and gaseous wastes. Due to waste generation from industries and mining activities, there is a great environmental concern and resource limitation. As these wastes can affect the environment proper planning is essential to manage the waste. As environmental problems impact public health & safety some practices such as reduction of generation and systematic waste disposal techniques should be followed. In view of associated environmental problems and their impacts on public health and safety, efforts must be made to minimize waste generation; systematic disposal practices must be followed. Hence, this study aims at a better understanding of spatial and temporal changes of unplanned dumping sites from 2006 to 2018. Google earth mapping is one of most advanced method for identification of rock solid waste clearly for collecting on satellite image data. This paper is centered on application of Google images in assessing the temporal changes of solid waste at limestone quarries in and around Yerraguntla. Time series multi date Google Earth Imageries of 2006 to 2018 are used to demarcate the evolutionary changes in Limestone waste disposal management and to understand the spatial and temporal changes that happened due to the changes by expanding the rock waste dump around Yerraguntla village, YSR district, A.P. Results revealed that the solid waste management in the study area is very poor which need to be properly monitored so as to mitigate the present and future environmental threats.

Keywords: Temporal changes, Solid waste, Limestone quarries, Google Earth, Yerraguntla;

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Introduction

Waste is defined as the discarded and discharged material during every stage of human life that leads to adverse impacts on human health and the environment (Bringi, 2007); Solid waste is now becoming a global problem in developing and also developed countries. Solid waste is a global environmental problem in today's world both in less developed and developed countries. Increasing population, rapid economic growth and the rise in community living standards accelerate solid waste generation in the world. At present waste management became a unavoidable challenge in a regional, national level and global level. Mining is one of the major incoming source for the developing countries like India, at the same time it's the major source of huge amounts of Solid waste generation in terms of overburden, milling, processing and tail

end wastes which need to be monitored and maintain proper mine plan and environmental policy particularly in opencast mining areas. The unplanned mining became a serious threat to the environment, resulting in the reduction of forest cover, erosion of soil in a greater scale, pollution of water, air and land and reduction in biodiversity. The problems of waste rock dumps become devastation to the landscape around mining areas (Sarma and Kushwaha, 2005). Mining activity, depending on the extent and the geographical location, causes some adverse impact on the surrounding environment with respect to human settlements, meteorological conditions, ambient air quality, water bodies, agricultural and forest land etc. In India limestone mines are .operated opencast mining method (IYMB, 2017). Comparatively, Opencast mining activity causes some adverse impacts on the surrounding environment, unless proper environmental management plan is adopted. Mine wastes require careful management to ensure the long-term stability of storage and disposal facilities, and to prevent and minimize air, water, and soil contamination. The inappropriate or unsafe management of wastes at mining operations continues to generate opposition from local communities, the general public, and non-government organizations, and has contributed to the negative public perception of the mining industry. It can be better monitored by the Remote sensing satellite data than the Underground mining. Selecting suitable sites for mining and also adopting all the guidelines prescribed by the Ministry of Environment Forests & Climate change (MoEF & CC) and Indian Bureau of Mines (IBM) can minimize the major possible impacts. With the present scenario where the extraction of minerals is done on a large scale it is became a highly matter of concern regarding the waste being generated by this mineral extraction, which has a severe impact on the environment, therefore a regular study should be taken in accordance with the application of latest technology in order to sustain this earth (Rajdeep Das 2013).

Study Area

Yerraguuntla is a Industrial village in YSR District in Andhra Pradesh. Geographically Yerraguuntla is located at 14.6333°N 78.5333°E with average elevation of 152 meters (501 feet). It is 37 km away from District head quarter Kadapa and well connected with rail Road and Sea ports. Cement Industries like India Cements, Zuari Cements and Bharati Cements etc., and has good number of stone polish industries, besides Rayalaseema Thermal Power Plant. Niduzivi, Koduru, Valasapalli areas of Yerraguuntla Mandal are known for Napa slabs/Kadapa slabs are black coloured limestone belonging to Koilakuntla Limestones (Sunitha et. al., 2014).YSR dist occupies lion share of limestone reserves in Andhra Pradesh, which is the major state among the limestone producing states with 313196000 tones production as per Indian Mineral year book 2017.

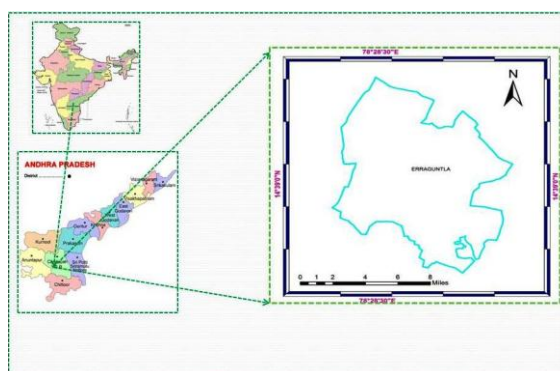


Figure 1: Location map of the study area

Methodology

A systematic integrated methodology adopted for the present study to assess the temporal changes in waste disposal. The intensive study of relevant literature, periodicals etc. has been carried out, based on the review of literature it is clear that remote sensing is the only effective tool to monitor the changes for multi date scenarios. Thus Google earth satellite images were chosen as the data source for the study area and the time series changes have been mapped for the years 2006 and 2018. Further detailed filed work carried out in the areas of waste disposal sites, mines, tailings, cement, stone cutting and polishing industries to know the types of waste

generation, handling, recycling and disposal. Data generated from the Google earth was compared and verified ground truth data by GPS survey, Topographic maps, field photographs, and questionnaire form local sources and mine and industrial management.

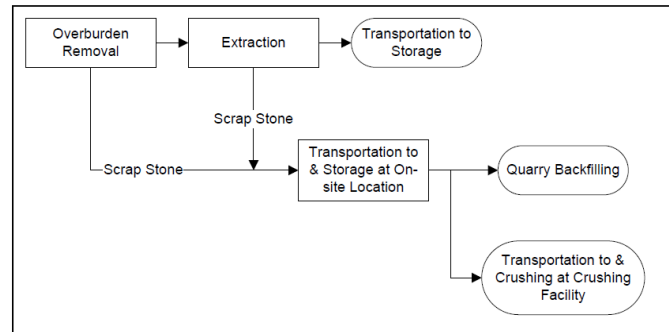


Figure 2: Flow chart of the limestone quarry processing.

Results and Discussions

Limestone mining waste disposal locations were identified on Google earth imagery, based on the image elements like tone, texture, shape, size, pattern and association, etc. The active dumps are clearly visible gray tone with medium to coarse texture due to the presence of Debris of waste and overburden and Irregular in shape and size. Whereas the mine pits with perfect shape and mainly in dark blue tone due to presence of water in the pits as shown in the figure (3&4).

In mining and cement industry, wastes are generated in every stage of the operations, in the present study we have identified two major locations of solid waste disposal, one is near the mine and second one is beside the road. Spatiotemporal changes in (2006 to 2018) as clearly depicted by drawing the red colored polygon for both the locations (Figure 3 A to F and Figure 4 A to F).

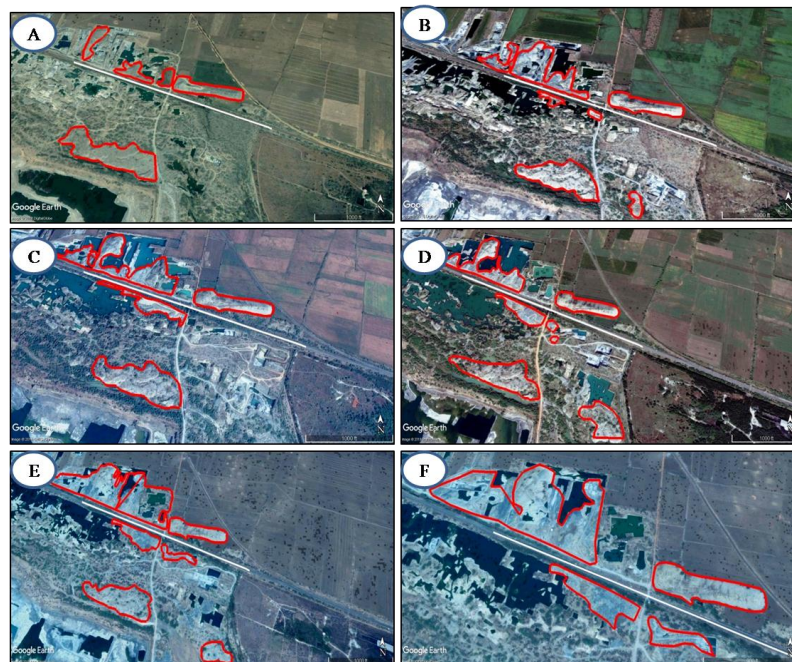


Figure 3: Temporal and Spatial Formation and Changes during 2006 - 2018 in location of solid waste beside railway track surrounding Yerraguntla village.

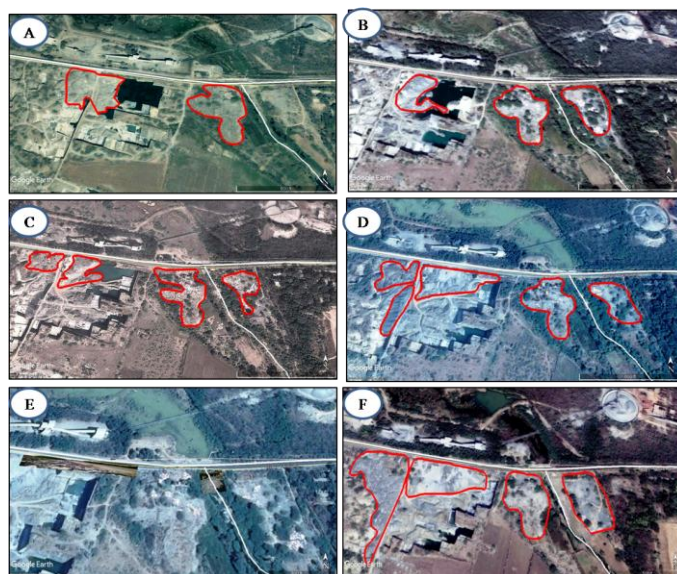


Figure 4: Temporal and Spatial Formation and Changes of solid waste during 2006 - 2018 beside highway road near India0 cement factory surrounding Yerraguntla village.

By observing the Google earth temporal images as shown on the figures (3&4) and the filed observations as shown in the figures 5, 6, 7 and 8 it is clear that overburden management and solid waste of these quarries are extremely poor; in most cases, the waste is just piled up in huge heaps, and mining companies do not bother themselves with measures to prevent run-off or fugitive dust from waste piles. The best use for overburden waste is to backfill the excavated land, but it is rarely done in practice as companies keep opening different faces of the mines without completely exhausting any one of them. The mines are characterized by large open pits surrounded by vast dump yards all around the croplands. Dust and slurry (figur 9 and 10) from these sites is being transported by the air and rainwater to nearby croplands ponds which is major threat to soil porosity, productivity, permeability and crop yield. During dry summer, these dumps become a key source of air pollution for the surrounding areas.



Figure 5: Limestone dumps at near India cement factory



Figure 6: Limestone at surrounding Yerraguntla village



Figure 7: Overburden of Niduzuvi Limestone quarry in a stream near the near India cement factory, Yerraguntla village



Figure 8: Solid wasteSurrounding the Yerraguntla village



Figure 9: water inundation in Limestone quarry fine dust during transport



Figure 10: Development of Surrounding Yerraguntla village surrounding Yerraguntla village

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Conclusion

The present study intended to assess the spatiotemporal changes in the disposal of solid waste derived from the different types and the quantities of mining waste, in and around Yerraguntla village area in Kadapa district by using the Google earth imagery. Based on the Satellite data analysis and ground truth verification it is concluded that over burden and the tailings from the limestone are extremely poor and there is no strict control on the solid waste disposal and management plan which resulted the dumping of waste in the nearby streams, road side and the banks of the highway. Fines from these dumps are carried by rainwater into nearby watercourses or lands and pollute both. It also observed that dust emissions from the mine and dumps become a key source of air pollution for the surrounding areas.

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